Workshop

AudioBox & ABControl: A platform for diffusion design and realisation

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The Richmond Sound Design AudioBox and Thirdmonk ABControl software provide an ideal platform for the design and presentation of multi-channel, multi-speaker electro-acoustic music diffusion.

The AudioBox is a hardware system which provides 8 channels of asynchronous hard disc audio playback and a 16 X 16 real-time DSP based matrix mixer. The matrix provides level control on all inputs and outputs and individual level control of each of the 256 crosspoints. In addition, delay and equalisation is available for all inputs and outputs. All level control parameters are dynamically adjustable in real-time providing the facility for complex spatial gestures and dynamic moves.

The ABControl software provides an intuitive user interface for the Audiobox. This software has been designed with electro-acoustic music diffusion in mind and has been employed on a number of electro-acoustic concert programs in Canada and elsewhere. ABControl provides facilities for diffusion cuelists, real-time diffusion control with movement capture, algorithmic diffusion generators, and vector based physical-speaker-independent speaker maps. Speaker maps allow for some degree of translation of a diffusion from one speaker set-up to another.

This workshop presentation will introduce the AudioBox and ABControl as a one-box solution for concert diffusion. The presentation will also consider the application of this system to soundscape installations, and sound effects and music playback for theatre. The workshop will address issues of performance, improvisation and compositional process. The related topics of human computer interaction, acoustics, psycho-acoustics and spatialisation will also be considered. Samples of various electro-acoustic compositions realised with the AudioBox will be presented during the workshop.